Flip-Flops:

- The memory elements used in "clocked sequential circuits" are called "flip-flops".
- It is a basic memory elements.
- These are binary cells and can store "one-bit of information".
- It has two outputs (Q and $\bar{Q}$), which are complemented to each other.
- It has two stable states which are known as the "1-state" and the "0-state".
- This Flip-flop circuit is also known as "Bistable Multivibrator" or Latch"
- A flip-flop circuit can be constructed from two NAND-gates or two NOR-gates.

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Direct-coupled RS Flip-flop or SR-Latch:

- It is a cross-coupled NAND or NOR-gates.
- The R and S are the first letters of the two input names.

NAND-gate Latch

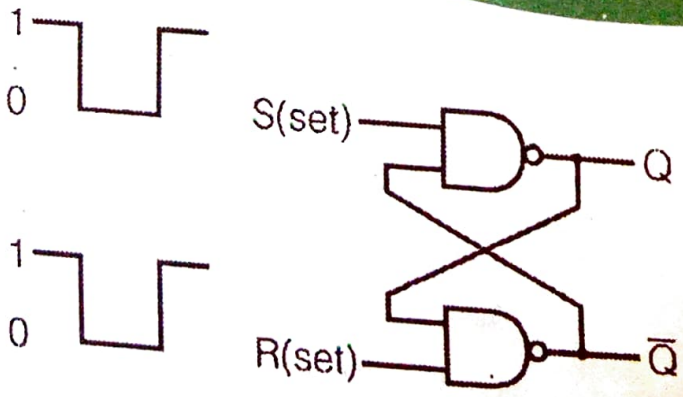


Fig. 6.2

- Truth table:

S	R	Q	$\bar{Q}$	
0	0	1	1	(Invalid)
0	1	1	0	
1	0	0	1	
1	1	0	1	(Previous state)
		1	0	

- After the set input returns to '1', a momentary '0' to the reset input causes a transition to the clear state.
- When both inputs go to '0' both outputs go to '1' a condition avoided in normal flip-flop operation.

■ NOR-gate Latch:

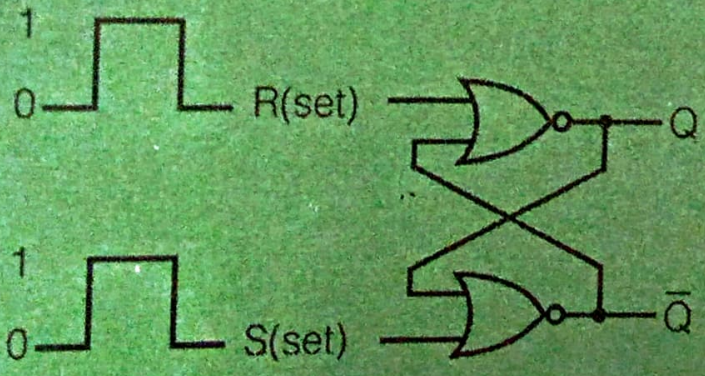


Fig. 6.3

- Truth table:

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S	R	Q	$\bar{Q}$	
0	0	0	1	Previous state
		1	0	
0	1	0	1	
1	0	1	0	
1	1	0	0	Invalid state

- The NOR gate latch operates exactly like the NAND latch except that the Set and Reset inputs are Active High rather than Active Low and the normal resiting state is $\boxed{S = R = 0}$.

Finally we get,

Inputs		Latch Gates	
S	R	NAND	NOR
		Q	Q
0	0	Invalid	No change or Previous
0	1	1	0
1	0	0	1
1	1	No change or Previous	Invalid

- Q will be set High by a High pulse on the Set input and it will be cleared Low by a High pulse on the Reset input.

o Clocked Flip-flops and Clock signals:

- Digital systems can operate either **Asynchronously** or **Synchronously**.
- In synchronous system, the exact time at which any output can change states are determined by a signal commonly known as the "Clock signals". The outputs can change states only when the clock makes a transition (also called edges).
- **Clock** changes from **0** to **1** → **Positive going transition (PGT)**.
- **Clock** changes from **1** to **0** → **Negative going transition (NGT)**.

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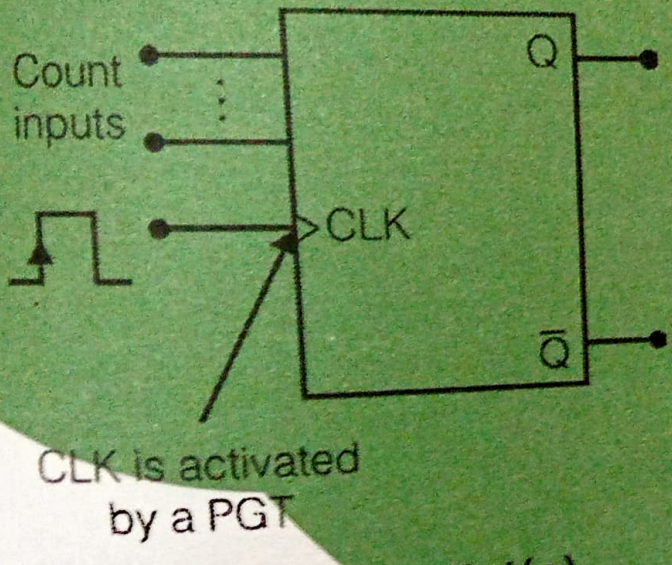


Fig. 6.4(a)

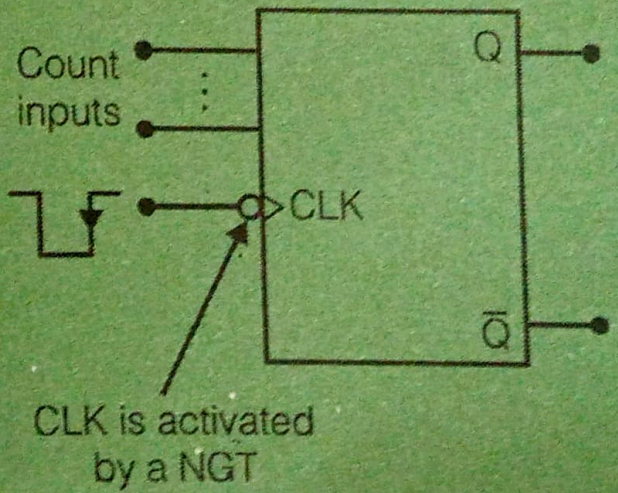


Fig. 6.4(b)

Clocked S-R Flip-flop:

- The clocked S-R flip-flop shown in figure 6.6(a) consists of a basic NOR Flip-flop and two AND-gates.

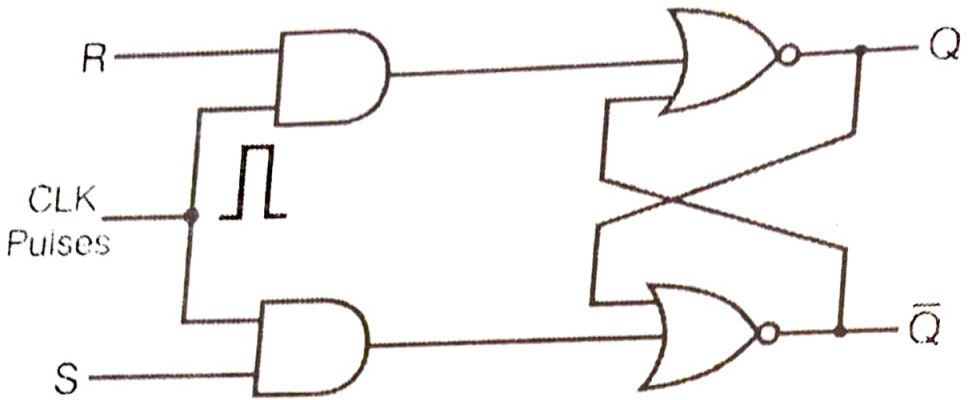


Fig. 6.6(a)

- *Truth table of S-R FF*

Clock	S_n	R_n	Q_{n+1}	
0	X	X	Q_n	
1	0	0	Q_n	→ HOLD state
1	0	1	0	→ RESET state
1	1	0	1	→ SET state
1	1	1	Invalid	→ FORBIDDEN state

⇒ Here S_n and R_n denotes the inputs and Q_n the output during the bit time '**n**'.

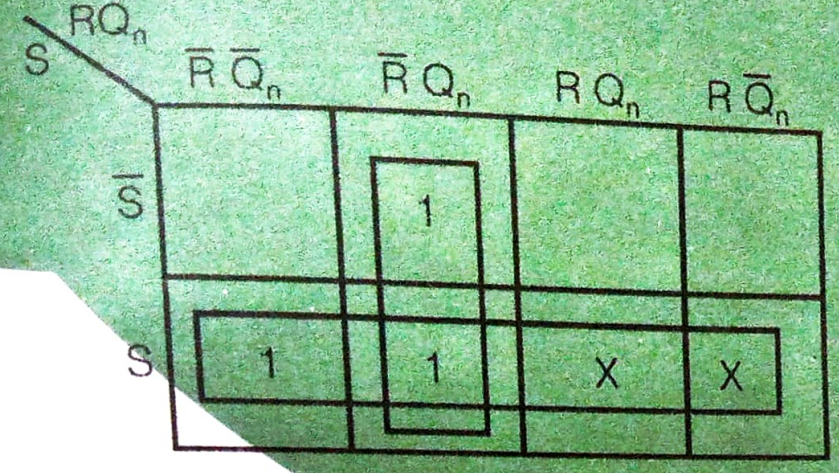
⇒ 'Q_{n+1}' denotes the output **Q** after CLK passes, i.e. in bit time $(n + 1)$.

Characteristice Table of SR-FF:

S	R	Q_n	Q_{n+1}	
0	0	0	0	→ Q_n
0	0	1	1	
0	1	0	0	→ 0
0	1	1	0	
1	0	0	1	→ 1
1	0	1	1	
1	1	0	X	→ Invalid
1	1	1	X	

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o K-Map:



∴ Characteristics equationof SR-FF is,

$$\boxed{Q_{n+1} = S + \bar{R}Q_n} \quad \ldots(6.1)$$

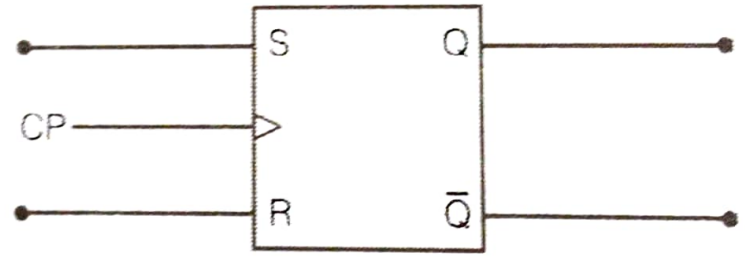
and

$$\boxed{\begin{array}{l} SR = 0 \\ SR \neq 1 \end{array}} \quad \ldots(6.2)$$

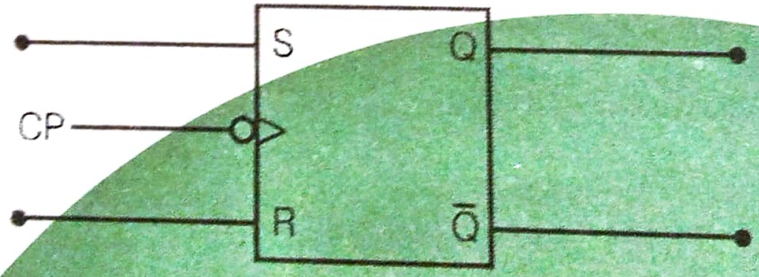
- The characteristic equation is an algebraic expression for the binary information of the characteristic table.
- It specifies the value of the next state as a function of the present state and the inputs.

Graphic Symbol:

- SR-FF has 3 inputs – S, R and CP



(+ve edge triggered SR-FF)



(–ve edge triggered SR-FF)

Fig. 6.6(b)

Disadvantages of SR-FF:

- Invalid states are present when both the inputs (S or R) made to HIGH.

  ⇒ To avoid this difficulty we have to use J-K-FF.

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❖ J-K Flip-Flop:

- A JK-FF is a refinement of the SR-FF in that the indeterminate (or invalid) state of the SR-type is defined in the JK-type.
- The data inputs are J and K which are ANDed with $\bar{Q}$ and Q respectively, to obtain S and R inputs i.e.,

$$\boxed{S = J\bar{Q} \text{ and } R = KQ} \qquad ...(6.3)$$

- This JK-FF is called an universal FF because the FFs like D-FFs, SR-FF and T-FF can be derived from it.
- Logic diagram:

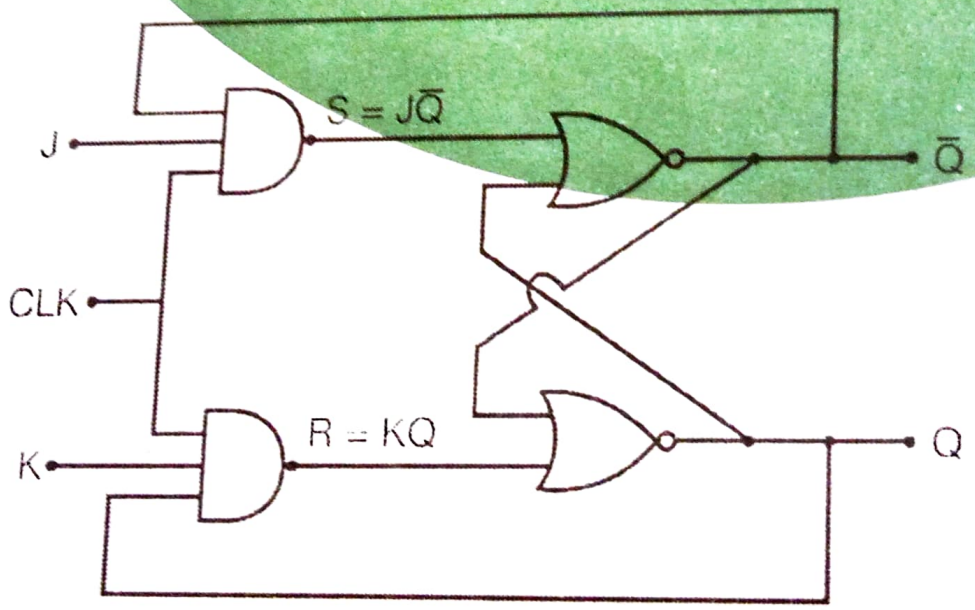


Fig. 6.7(a)

- Graphical diagram:

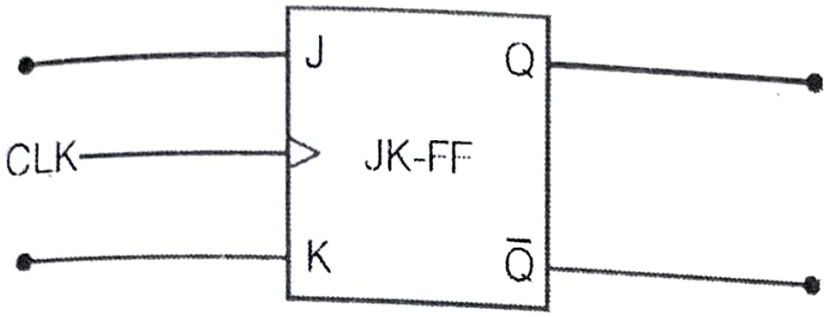


Fig. 6.7(b)

- Truth Table of JK-FF

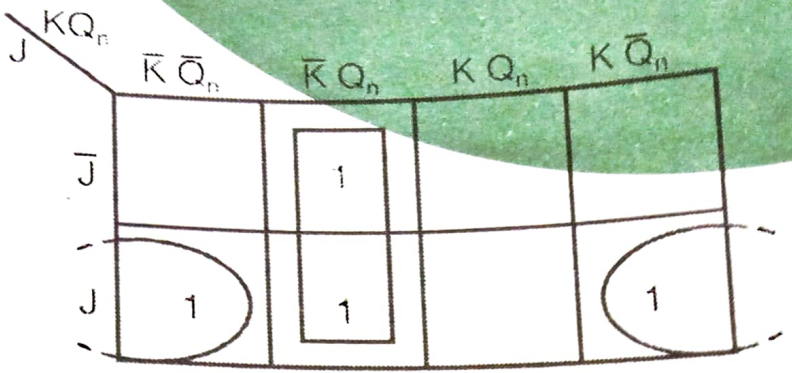
Clock	J	K	Q_{n+1}	
0	X	X	Q_n	
1	0	0	Q_n	→ HOLD state
1	0	1	0	→ RESET state
1	1	0	1	→ SET state
1	1	1	$\overline{Q}_n$	→ TOGGLE state

o Characteristic Table:

J	K	Q_n	Q_{n+1}	
0	0	0	0	→ Q_n
0	0	1	1	
0	1	0	0	→ 0
0	1	1	0	
1	0	0	1	→ 1
1	0	1	1	
1	1	0	1	→ $\overline{Q}_n$
1	1	1	0	

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- K-Map:



∴ Characteristic equation is,

$$Q_{n+1} = J\bar{Q}_n + \bar{K}Q_n \qquad \ldots(6.4)$$

[IES-2007, 2006, 2003]

- Disadvantage of JK-FF:
  - Race-Around Condition
    - The difficulty of both inputs '1' i.e. S = R = 1 being not allowed in S-R FF is eliminated in a JK-FF by using the Feedback connections from output to the input.
- When J = K = 1 and Q = 0, and a CLK as given in figure 6.8 is applied at the CLK input. After a time "$t_{pd(FF)}$" (Propagation delay through two NAND gates in series), the output will change to Q = 1, after another "$t_{pd\ (FF)}$" output will change back to Q = 0.
- For the duration "t_{pw}" of the CP, the 'Q' will oscillate back and forth between 0 and 1. At the end of the CP, the value of 'Q' is uncertain. This situation is referred to as the "Race-around condition".